Statistical Physics for ML

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Contents

1 To-do

- No need to add lots of details, since we're not writing a book. Just high-level notes.
- However, there are some gaps in our understanding (see the "?" symbol below). [SJ: For me personally, the distinction between "replica symmetry ansatz", "no overlap gap", and "landscape complexity" is quite hazy. Also, my understanding of the Optimization and Sampling sections is weak...]
- It would be good to add more examples, especially if we present this to the reading group. Specifically, phase transitions in statistical learning (lasso, GLMs) and combinatorial optimization problems (CSPs, perceptron).
- There are many adjacent topics not listed below: correlation inequalities, exponential random graph models (ERGMs), graph limits, large deviations, superconcentration, noise sensitivity, etc. We can also add these to our toolbox.

2 Background

Statistical physics studies systems consisting of very many interacting parts. Systems are either in equilibrium ("statics") or out of equilibrium ("dynamics"). We will primarily be interested in the former, though dynamics is related to optimization and diffusion processes.

Some of the most well-known equilibrium statistical physics models are models of ferromagnets, which can be thought of as a large number of interacting electrons, each having a tiny magnetic spin. A configuration is a certain arrangement of these spins, and each configuration has a certain energy, given by the Hamiltonian function. At a given temperature T, the configuration of the system is modeled as a random sample from a probability distribution called a Gibbs measure. The Gibbs measure associated to the Hamiltonian H at temperature T is the probability distribution over configurations proportional to $\exp(-\beta H(S))$, where $\beta = 1/T$ is the inverse temperature. From this definition it is clear that lower energy configurations are given higher probability, which is completely natural.

Of course, as macroscopic beings, we can't observe the microscopic properties of the realized configuration; we only measure large-scale quantities in the laboratory. Such a quantity is called an observable. Examples of observables include the average magnetization m (defined as the average value of any given spin), the free energy, and so on. It is often the case that observables are "self-averaging", i.e. they concentrate around their expectation as the size of the system "N" is taken to infinity. The limit "N to infinity" is called the thermodynamic limit in physics, because only in this limit do we recover macroscopically-meaningful properties. Thus we only care about these models in the thermodynamic limit. We discuss the technical difficulties of defining such "infinite volume Gibbs measures" below.

At this point we haven't yet written down an example of a Hamiltonian, which models the interactions in the system. In all of our examples, the Hamiltonian will be a certain (random) quadratic form in the configuration S. The coefficients of the degree 2 terms will be called "coupling constants". (In more complicated statistical physics models, higher-order interactions can be considered.) Broadly speaking there are two kinds of interactions in statistical physics: mean field and non-mean field. In a mean field model, all pairs of spins interact in exactly the same manner. If one were to draw an "interaction graph", it would be the complete graph on N vertices. For ferromagnetism, the canonical mean field model is the Curie-Weiss model, where spins can either be $+1$ or -1, and whose Hamiltonian is

$$H(S) = -\frac{1}{2N}\sum_{i,j} S_i S_j - h\sum_i S_i.$$

The canonical non-mean field model is the Ising model of ferromagnetism. Here the spins are placed at the vertices of the integer lattice $\mathbb{Z}^d$, and each spin only interacts with its nearest neighbors. The Hamiltonian is the same as for C-W, except the sum only runs over adjacent pairs i and j. Non-mean field models are generally harder to analyze. (Here "h" denotes an external magnetic field that interacts with each individual spin. Spins are obviously encouraged to align with h.) (The distinction is not as sharp as it may seem: as it turns out, when d is taken to infinity, the Ising model begins to behave like a mean field model.) There is another distinction to be made: ordered versus disordered systems. In the latter case, the coupling constants are random, which induces a random Gibbs measure; more on that below.

Our goal is a complete understanding of the (infinite volume) Gibbs measure(s) at every inverse temperature β and every external field h. Of course, the difficulty is that the spins are dependent in a highly complicated manner. (It may seem strange to write "measure(s)". Haven't we defined a single measure for each pair of β and h? We will clarify below.) The most important phenomena to understand are phase transitions in the Gibbs measure as β is varied. From playing with bar magnets, we know that the magnetization goes to zero if one increases the temperature past a certain point, called the Curie temperature. Can this phase transition be proven mathematically? Consider the C-W or Ising model. It's obvious that if β is taken to zero (i.e. if we take the "high temperature limit"), then the Gibbs measure degenerates into the uniform distribution, and all spins behave independently. In such a setting it is obvious that the magnetization should be zero. On the other hand, if β is taken to infinity (i.e. if we take the "low temperature limit"), then the interaction strength is very strong, and the Gibbs measure concentrates on its ground states (the lowest energy configurations). At zero external field, this is the average of two point masses at the "all $+1$" configuration and the "all -1" configuration. Thus if one were to approach $h = 0$ from the positive direction for large β, then one would be left with a positive residual magnetization (and this is of course how we prepare ferromagnets). This is a first glimpse of the phase transition. A precise definition of the phase transition requires the language of infinite volume Gibbs measures. After all, for a fixed finite N, no sharp phase transition in β will occur; all relevant observables (such as the free energy) will be analytic functions of β and h. Fundamentally, a phase transition corresponds to non-uniqueness of an infinite volume Gibbs measure at a given pair (β, h). In both the C-W model and the Ising model for $d > 1$, there will be a unique infinite volume Gibbs measure for all pairs with $h \neq 0$ and for all pairs $(\beta, 0)$ with $\beta < \beta_c$. However, along the "coexistence line" $(\beta, 0)$ for $\beta > \beta_c$, suddenly there will exist at least two distinct infinite volume Gibbs measures at $(\beta, 0)$. In fact, because the set of infinite volume Gibbs measures is convex, there will be infinitely many Gibbs measures, obtained by convex combinations of extremal Gibbs measures.

At this point you may have many questions. We haven't even written down an infinite volume Gibbs measure; on what space does it live? What is its density? How does it relate to the finite volume Gibbs measures at (β, h) for each N? If we think of the infinite volume Gibbs measure as a "limit" of its finite volume counterparts, does that mean a single sequence of measures can have two limits? First of all, an infinite volume Gibbs measure mu lives on the countably infinite hypercube $\{\pm 1\}^{\mathbb{N}}$. It's not meaningful to write down a density, so we won't. Instead, let us postulate certain abstract finite volume "compatibility conditions" that mu should satisfy. The idea is, if we look at any finite subset U of the infinite hypercube (say of size N), then the marginal distribution of mu on these sites should equal the corresponding finite volume Gibbs measure on N spins, with a certain natural boundary condition induced by U complement. Formally, these conditions are given by the Dobrushin-Lanford-Ruelle equations (DLR equations). This approach might remind you of Kolmogorov's extension theorem, which in probability theory is used to define countably infinite product measures by enforcing compatibility conditions on the marginals. The finite and infinite volume Gibbs measures are certainly not product measures, but the idea is the same. (One can also define infinite volume Gibbs measures either by directly taking limits of finite volume Gibbs measures.)

Now we present intuition for why two distinct infinite volume Gibbs measures exist at

$(\beta, 0)$ for $\beta > \beta_c$ in these models of ferromagnetism. Recall the average magnetization m. It's obvious that $m > 0$ for the unique infinite volume Gibbs measure at (β, h) for $h > 0$, and vice versa for $h < 0$. (Simply compare each configuration S with the configuration obtained by flipping all of its spins. The one with more sign(h)'s will have a strictly higher probability.) Consider taking a limit of infinite volume Gibbs measures at (β, h) for a sequence of h's going to zero from above. Intuitively, this should yield an infinite volume Gibbs measure at $(\beta, 0)$ whose average magnetization equals the limit of $m(\beta, h)$ as h approaches zero from above. Similarly, if we take a limit as h approaches zero from below, we should obtain an infinite volume Gibbs measure at $(\beta, 0)$ whose average magnetization equals the limit of $m(\beta, h)$ as h approaches zero from below. The result states that at $\beta < \beta_c$, the resulting Gibbs measures are identical, but for $\beta > \beta_c$, the two limiting Gibbs measures are actually distinct. In other words, the magnetization function $m(\beta, h)$ has a jump discontinuity along the coexistence line, but $m(\beta, h)$ is continuous everywhere else in the (β, h) plane. So to show the existence of two distinct Gibbs measures at $(\beta, 0)$ for $\beta > \beta_c$, we just need to show that the limiting magnetizations (in physical terms, the "residual magnetizations") are distinct (i.e. they are not both zero). One can also reformulate this observation as a differentiability property of the free energy. By basic calculus, $m(\beta, h)$ equals the partial derivative of $\Phi(\beta, h)$ with respect to the external field h. Thus the phase transition corresponds to a lack of differentiability of Φ along the coexistence line: its left and right partial derivatives in h don't agree along the ray. (In my mind, I imagine Φ looking like the absolute value function along the coexistence line.)

A complete understanding of the Gibbs measure is completely out of reach for disordered non-mean field models such as the Edwards-Anderson model, but over the years great strides have been made in both the disordered mean field and the ordered non-mean field settings. Of particular relevance to statistical inference are disordered mean field models.

You may wonder why the Gibbs measure is defined the way it is: an exponential tilt of a given base measure. This choice is not at all arbitrary, and it relates to the concept of "equivalence of ensembles". It also explains why calculating the free energy is so crucial to understanding the Gibbs measure. In statistical physics, the Gibbs measure is also called the canonical ensemble. It describes a system at a fixed temperature. In the laboratory, it's often reasonable to assume a fixed T, because our system is typically in thermal equilibrium with a fixed-temperature heat bath. However, there is a more fundamental theoretical model known as the microcanonical ensemble. In the microcanonical ensemble, we instead fix the internal energy of the system, and postulate that all configurations at that energy are equally likely. In other words, given an internal energy E, we sample a configuration uniformly at random from the set of all configurations with $H(S) = E$. The microcanonical ensemble is a highly idealized model, because it assumes the system is completely isolated from the rest of the world, so that its energy does not change. It's obvious that conditional on $H(S) = E$, the canonical ensemble reduces to the microcanonical ensemble. However, apriori in the canonical ensemble, any energy E can be attained. The magic of the Gibbs measure is that aposteriori, the internal energy concentrates around a single value in the thermodynamic limit. In other words, for large N, the canonical ensemble effectively reduces to the microcanonical ensemble, at a very special value of E. What is this value of E? It turns out to be nothing other than the free energy Φ. The proof of this fact is a simple but profound computation, due to Boltzmann. The Gibbs measure tells us the distribution over (N-dimensional) configurations S, but we seek the distribution of the (scalar) Hamiltonian $H(S)$ itself. Writing $N(E)$ for the number of configurations that attain the internal energy $H(S) = E$, the probability that $H(S) = E$ is proportional to the product $N(E)\exp(-\beta E)$. Here we begin to see the key optimization problem at play, referred to as "the competition between entropy and energy". As E begins to increase, the multiplicity of states $N(E)$ increases; but at the same time, higher energy is penalized by the Gibbs measure, and $\exp(-\beta E)$ decreases. Taking logs, we seek the argmax in E of $(\log N(E) - \beta E)$, which is a Legendre transform that can be computed for specific models.

Gibbs measures appear in natural statistical problems, and this is the fundamental reason why we're learning these statistical physics techniques. Perhaps the simplest example is the $\mathbb{Z}_2$ synchronization problem, where we seek to infer a vector v living on the hypercube from a noisy observation Y of the rank 1 matrix vv^T. The specific model we consider is

$$Y = \frac{\lambda}{n} vv^T + W,$$

where W is a $GOE(n)$ matrix and λ indicates the SNR. Let's take a Bayesian perspective and put a uniform prior on v. Then the most natural estimator is the Bayes optimal estimator of v, given by the posterior mean of v given the observed Y. If you write down the posterior distribution, you find $p(u|Y)$ is proportional to

$$\exp\left(\left|Y - \frac{\lambda}{n} uu^T\right|^2\right).$$

If you plug in the formula for Y, this simplifies to

$$p(u|Y) \propto \exp\left(\frac{1}{2}\lambda u^T W u + \frac{1}{2n}\lambda^2 u^T 1\right).$$

Now the expression in the exponent is a quadratic function of u, so we have a (random) Gibbs measure! In fact, this is a tilted version of the Sherrington-Kirkpatrick measure arising in the theory of spin glasses.

Two key differences between the C-W model and the S-K model are that (1) the latter is a random Gibbs measure, and (2) the latter is disordered. (1) is self explanatory: the coupling constants are given by the random matrix W. (2) requires explanation. It's clear that W contains both positive and negative numbers with high probability, whereas all coupling constants in the C-W model are $+1$. Whereas each spin in the C-W model experiences the same environment (one that encourages agreement between spins), this is not the case for the S-K model. Each spin in the S-K model has a different environment. Some spins may want to disagree with their neighbors, while others may want to agree. These competing interests cannot all be satisfied, and so the landscape of ground states is much more complex than the C-W model. The term for this phenomenon is "frustration."

(From Sourav Chatterjee's talk at Harvard on YouTube:) So far we haven't made a precise definition of what a "spin glass phase" is. Two defining properties of this phase (in contrast to the "ferromagnetic phase" of the C-W/Ising model) are (1) multiple ground states ("multiple valleys") and (2) chaos with respect to the couplings. For (1), the physics intuition is that when you flip all the spins in a given box of the ground state, due to the CLT, the energy doesn't change by much, so you get another near-ground state. Thus we expect a bunch of nearby ground states in the landscape. This is in stark contrast to the ferromagnetic case. For (2), it means that slightly perturbing (either through interpolation or resampling) the couplings can cause the sampled configuration to dramatically change. Specifically, the new configuration will have almost 0 overlap with the original configuration with high probability.

For technical details on the foundations of statistical mechanics, see the mathematical physics textbooks Friedli and Velenik (2017) and Bovier (2006).

3 Phase transition in magnetization

This was touched on in the background section. Here we derive the Ising magnetization (at zero external field) as a function of β under the mean-field assumption. This assumption is valid for large d. To solve the Ising model in 2-d without approximation, you need a certain algebraic device called Onsager's transfer matrices.

(Leonard Susskind's Modern Physics Statistical Mechanics Lecture 8 on YouTube:) Assume every spin has the same average magnetization m; this is the origin on the term "mean-field." Let's focus on a single spin σ_i and treat it as a system in itself. Let's replace every other spin σ_j with its average m. Then the partition function of the subsystem is

$$Z(\beta) = \exp(+2d\beta m) + \exp(-2d\beta m) = 2\cosh(2d\beta m).$$

Thus the free entropy is

$$\Phi(\beta) = -\log 2\cos(2d\beta m).$$

Now note that the derivative $\partial\Phi/\partial\beta$ equals the average energy of spin i in this subsystem, so that

$$\text{average energy of } i = 2dm \tanh\tan(2d\beta m).$$

(This thermodynamic relation $\partial\Phi/\partial\beta = U$ is distinct from the free energy trick introduced below.) But in this one-spin subsystem, the average energy is precisely $2d\beta$ times the magnetization $\langle\sigma_i\rangle$! We deduce that

$$\langle\sigma_i\rangle = \tanh(2d\beta m),$$

so by our mean-field assumption, we recover the self-consistent equation that characterizes $m(\beta)$:

$$m = \tanh(2d\beta m).$$

This will have only the solution $m = 0$ for low β, but for high β, there will be three intersections at m^*, $-m^*$, and 0. We can explicitly solve for the critical β_c by setting the slope of the right-hand side equal to 1. We find $2d\beta_c = 1$, or equivalently $\beta_c = \frac{1}{2d}$. This implies that the critical Curie temperature increases linearly with d. This is also reminiscent of the critical point in percolation on regular trees.

This technique is essentially a simplified cavity method; study a subsystem that experiences a "cavity field" from the remaining spins.

The concept of spontaneous symmetry breaking: since there is zero external field, the Hamiltonian $H(\sigma)$ is symmetric with respect to a global sign change. Above β_c, two dominant configurations emerge (mostly $+1$ vs mostly -1), and these are equally likely. In real life, the system must make a choice between these two options; it must break the symmetry. This is usually caused by a tiny stray external magnetic field that biases the system to align with it.

Imagine moving in (β, h)-space and crossing the critical line. A typical configuration goes from "a sea of -1's" when $h < 0$ to "a sea of $+1$'s" when $h > 0$; this is a jump discontinuity known as a "first-order phase transition." If you circumvent the critical line by taking a detour to the left of β_c, then you avoid this jump discontinuity. Instead, you watch "a sea of -1's" continuously morph into "a mixture of -1's and $+1$'s" and ultimately "a sea of $+1$'s." It's quite analogous to the phase transition from gas to liquid.

4 Free energy trick

The partition function is a sum over 2^n states, and so is exponentially large in n. The free entropy is $-\log Z$, minus the exponent of the partition function. The free energy is $-\frac{1}{\beta}\log Z$. It turns out that the free energy characterizes the Gibbs measure. This is familiar from the theory of exponential families in statistics. In statistical physics, this is known as the free energy trick: the expected value of any observable can be computed by taking a derivative of the free energy of a perturbed system. Thus physicists have developed many methods for estimating the free energy of a Gibbs measure: the cavity method, the replica method, the n-step replica symmetry breaking hierarchy, the Bethe free energy, the mean-field free energy, the TAP free energy, and so on.

One interesting fact about the free energy (which for a given model is just a single number, not a function) is that it often is expressed in a variational form. This was essentially mentioned above in the discussion of the competition between entropy and energy. The free energy (or equivalently, the partition function) involves a sum over 2^n states. This can be rewritten as a sum over all energies of the number of states with the given energy times the Boltzmann factor. It turns out that the number of states (as a function of energy) is sharply peaked at a single point, so essentially only one term contributes to this sum. Thus we simply approximate the entire free energy by replacing the sum with the maximum of the summand, and we get a variational representation for the free energy with respect to some optimization variable. You can think of this as a saddlepoint approximation. Again, to emphasize, the optimization variable isn't an observable of the system; it's just a dummy variable. Expectations of observables only have one value, they don't vary.

In simple models like the C-W model, the optimization objective can be handled exactly, because counting the number of configurations with a given energy is easy. But this gets harder when you're considering more complicated models, or when you're considering replicated systems. In these cases, you avoid combinatorics altogether by using "field-theoretic" techniques. In essence, you "count" by integrating a delta function, and then you formally apply Fourier inversion to this delta function. More on that in the section on the replica method.

5 Cavity method

The key idea is to relate the system with N spins to the system with $(N+1)$ spins. If you can get a relation of the form

$$H_{N+1} = H_N + (S_0 \text{ dependent terms}),$$

then you're well on your way to computing key observables. Take the exponential of $-\beta$ times both sides, take the sum/expectation with respect to the Gibbs measure on $(N+1)$ spins, rewrite the right-hand side as a sum/expectation with respect to the Gibbs measure on N spins, and you will obtain a self-consistent equation either for the average magnetization m^* or the limiting free energy per spin. The cavity method essentially studies the marginal distribution of S_0 conditional on spins $S_1, \ldots, S_n$, and then removes the conditioning with respect to the Gibbs measure on N spins. The conditional marginal distribution of S_0 is very simple: S_0 interacts with nothing but an effective external field, given by $(\bar{S} + h)$. This effective external field is also called the cavity field.

(The cavity method is also reminiscent of the pseudolikelihood in mathematical statistics. The PL is also similar to mean-field approximations, except it's not a product measure.)

6 Replica method

This technique is far less rigorous than the cavity method, and is rather bizarre at first glance. However, its predictions have been made rigorous in certain cases. When we use the replica method, we are implicitly assuming a certain structure on the Gibbs measure:

that the probability mass is mostly contained in a single "cluster". In Zdeborova notes, the replica method is used to study RFIM (actually, the RFCW model?).

Here is what replica symmetry "really means." We say that a Gibbs measure is replica symmetric if, given two iid samples $\sigma^{(1)}, \sigma^{(2)}$ (experiencing the same "quenched" randomness), the overlap $\langle \sigma^{(1)}, \sigma^{(2)} \rangle / n$ concentrates. In other words, the Gibbs measure is supported on a single contiguous cluster. k-step replica symmetry breaking occurs when the overlap concentrates around k different values. The overlap gap property is a certain type of replica symmetry breaking, where the pdf of the overlap (on $[0, 1]$) is disconnected, with a subinterval of zero density. When the OGP holds, statistical-computational gaps emerge: the most computationally efficient procedure might not attain the minimax bound. The k-step RSB assumption manifests as an ansatz when solving the first-order conditions of the saddlepoint approximation. Now back to the details.

To compute the annealed free energy $\mathbb{E}[\log Z]$ (where the average is taken over the random field h_i), we reduce to a moment calculation via

$$\mathbb{E}[\log Z] = \lim_{n \to 0} \frac{\mathbb{E}[Z^n] - 1}{n}.$$

We compute $\mathbb{E}[Z^n]$ for each positive integer n, and assume that the resulting formula can be analytically continued to the whole positive real line.

The raw n-th moment requires a formidable combinatorial calculation. To avoid combinatorics, we use two field-theoretic tricks: for each replica $a \in [n]$, introduce dual variables m_a and $\hat{m}_a$. (m_a can loosely be thought of as the average magnetization of replica a, and $\hat{m}_a$ is a frequency variable; but again, we emphasize that these are just optimization variables!) The n-th moment is rewritten as a double contour integral of an exponential function of m_a and $\hat{m}_a$.

To apply the saddlepoint approximation (the multivariate analogue of the Laplace approximation), we would need to optimize the exponent, a function of $2n^2$ variables. This is where we use the replica symmetry ansatz: we assume the optimum is achieved at a point where all m_a's are equal, and where all $\hat{m}_a$'s are equal. (For intuition, see the discussion of the "structure" of the Gibbs measure above.)

This reduces the optimization to a bivariate problem, which is easy. Eliminating $\hat{m}$, we eventually recover a variational formula

$$\Phi(\beta, h) = \operatorname{argmax}_m \phi(m) = \phi(m^*)$$

for some replica symmetric potential ϕ. As usual, the magnetization m^* obeys a self-consistent equation that is very similar to the deterministic field setting.

Guerra's interpolation technique can be used to rigorize the replica predictions. But as discussed below, we are not always in a replica symmetric regime.

7 Replica symmetry breaking

If instead there are multiple far-apart clusters (so that the measure is somewhat "bimodal"), then the replica-symmetric ansatz is broken, and we must turn to more complex versions of the replica method, known as the n-step replica symmetry breaking (n-RSB) hierarchy. Roughly speaking, the n-RSB ansatz assumes that the Gibbs measure consists of n well-separated clusters. For extremely complex Gibbs measures, one might even need infinity-RSB; this is what Parisi used to compute the exact formula for the free energy of the S-K model for all (β, h). (His paper Parisi (1979) is somehow only 3 pages long...) Talagrand's monumental proof of the Parisi formula for the S-K model makes this rigorous.

The precise definition of n-RSB has to do with the distribution of the replica overlaps, i.e., the inner products $\langle S^{(a)}, S^{(b)} \rangle$ for all $a, b \in [n]$. If these concentrate around $(n+1)$ distinct values, then we have n-RSB. Here $\langle S^{(a)}, S^{(b)} \rangle$ is defined as

$$\frac{1}{N} \sum_i S_i^{(a)} S_i^{(b)},$$

which is related to the number of agreeing spins.

(From Sourav Chatterjee's CLAPEM talk on YouTube:) Physicists believe that the distribution of overlaps in spin glasses is characterized by the "hierarchical structure of pure states." They believe pure states can be hierarchically clustered; think of the clusters as the vertices of a tree. The limiting overlap between two pure states is determined by the smallest cluster in which both are present. If this tree only has finite depth, then we have k-RSB for some finite k. Otherwise, we have infinity-RSB, and the overlap distribution is continuous. This tree has to do with the ultrametricity property... [SJ: This doesn't make sense to me; what is a "pure state"? Is it simply a deterministic infinite volume Gibbs measure? And do we define the overlap as the expected overlap for replicas drawn from each?...]

8 Belief propagation

How do we efficiently compute the mean of a Gibbs measure? Equivalently, how do we compute marginal distributions at each spin? Equivalently, how do we compute the partition function of subgraphs? The origin of the BP algorithm is an exact recursion that solves these problems on trees. The derivation is pretty straightforward. You need the notion of factor graphs, a visual depiction of a Gibbs measure. First you do the partition function recursion, making use of the fact that "a sum of products is a product of sums." Then you do some normalization to get the recursion involving the marginal distributions; recall that the probability that site i has spin σ_i is just the partition function conditional on site i having spin σ_i divided by the total partition function over all assignments. These recursions can be interpreted as "message passing" procedures: each factor node a passes a message to each adjacent variable node i telling it what it believes i's marginal distribution is, based on inputs from variables $j \in \partial a \setminus i$; and each variable node i passes a message to each adjacent factor node a telling it what it believes its own marginal distribution is, based on inputs from factors $b \in \partial i \setminus a$. The partition function approximation is called the Bethe free energy ansatz. One can show that if our factor graph is locally tree-like (in that it has no short cycles), then BP asymptotically yields the right results. You run BP until it converges, and hope that you've reached the right BP fixed point. (There can be multiple, and you can analytically solve for them using certain ansatzes on the structure of the marginal distributions.) For details, see Chapters 4 and 5 of Krzakala and Zdeborová (2022).

9 Approximate message passing

When computing marginal distributions, BP is computationally heavy, because you're passing around a lot of "beliefs"/marginal distributions. AMP is an approximation that only passes around mean-variance pairs. If we're trying to estimate the Gibbs mean using the marginals, then AMP allows you to fully understand the distribution of the estimator at each time step.

AMP is not the naive Gaussian approximation to the messages, but involves a subtle Onsager correction. (This might not make sense in the case of spin systems on $\{\pm 1\}^n$, but it makes sense when our Gibbs measure is supported on $\mathbb{R}^n$, and when we have a dense factor graph. See Mei (2021) for the Gibbs measure defined by the lasso objective.)

It turns out AMP iterates can be described by a certain low-dimensional iteration called state evolution. This captures the effective noise level in the estimates at each step. That's why AMP results are often phrased as "the estimate is distributed as the true parameter plus $\eta_t G$, where $G \sim N(0, 1)$ is independent of the true parameter and η_t is the state evolution parameter at step t."

AMP can also be derived from a pure optimization perspective, as an improvement to ISTA. The difference is exactly the Onsager correction term. Again lasso is a good example. See Montanari et al. (2012).

The true origins of AMP lie in computing the expected magnetization m^* of the SK model. As we discuss below, there is an approximation of the free energy called the TAP free energy, given in variational form. It turns out that the TAP free energy obeys a fixed-point equation as a function of the expected magnetization m^*, so a natural iteration scheme can be set-up to compute it. This leads to AMP.

10 Random constraint satisfaction problems

These include solving random k-SAT instances, finding k-colorings of sparse random graphs, finding independent sets of a given size in sparse random graphs, and so on. We are interested in phase transitions in the structure of the space of solutions as the constraint density α increases. Conjecturally, we expect that once the solution space shatters into exponentially many clusters, it becomes computationally hard to find a solution.

11 TAP free energy

Consider the problem of computing the mean of a high-dimensional Gibbs measure $p(\sigma)$ on the hypercube $\{\pm 1\}^n$. (We already considered this problem in the BP section.) Why is this interesting statistically? Because in some Bayesian estimation problems, the posterior happens to be a Gibbs measure, and we want to compute some expectation with respect to the posterior. For $\mathbb{Z}_2$ synchronization, the Bayes optimal estimator of the rank-1 perturbation is the expected value of $\sigma\sigma^T$ with respect to the posterior. If m is the mean of the posterior, we can hope that mm^T is a good approximation to the Bayes optimal estimator.

So back to our problem of computing the mean. This is a high-dimensional integral, which is hard. A technique from Bayesian inference called variational inference offers a way out. Since we can't compute the mean of the true measure, compute the mean of some approximation. As a first step, we could perform mean-field variational inference, where we approximate $p(\sigma)$ by some product measure $q(\sigma)$ given by the product $\prod_{i=1}^n q_i(\sigma_i)$. Let's pick the product measure that's closest to $p(\sigma)$ in KL divergence, and then compute its mean, which is a piece of cake. When do we expect this approximation to be accurate? Precisely when the spin-spin correlations in the Gibbs measure are weak, i.e. generally at high temperature, or low β. If $q^*(\sigma)$ is the minimizer, then since each $q_i^*(\sigma_i)$ is a distribution on the set $\{\pm 1\}$, the approximate mean equals $(q_i^*(+1) - q_i^*(-1))_{i\in[n]}$.

Let's now rewrite the variational inference objective slightly. The quantity $D_{KL}(q\,||\,p)$ is defined as $\mathbb{E}_q\left[\log \frac{q(\sigma)}{p(\sigma)}\right]$. If we rewrite this, we get $(F_{mf}(q) + \log Z)$, where $\log Z$ is the log-partition function of the true Gibbs measure p.

Since the set of product measures q on the hypercube is in bijection to the set $m = (m_i)_{i\in[n]}$ of means of the marginals $q_i(\sigma_i)$, we can also treat this as the minimization of $(F_{mf}(m) + \log Z)$ over m.

Assuming that the mean-field approximation is accurate, this quantity should be close to zero at the optimum, and so the optimized expression $F_{mf}(m^*)$ should be a good approximation of the free entropy. This is called the "mean-field free entropy" (or, with correct normalization, the "mean-field free energy").

Thus, if this approximation is valid, the free energy is given by solving the variational problem

$$\min_m F_{mf}(m).$$

But what if we're at low temperature, and there are strong spin-spin correlations? In this case the mean-field approximation breaks down. In the case of the SK model, physicists Thouless, Anderson, and Palmer proposed a modification to the mean-field free energy in the 70s, called the TAP free energy. The only difference between F_{mf} and F_{TAP} is the subtraction of the so-called "Onsager correction."

The derivation of F_{TAP} is actually very simple. You take the true free energy, introduce a new parameter α in front of the pairwise interactions in the SK Hamiltonian, do a Taylor expansion in α, keep up to second-order terms, and set $\alpha = 1$. This is known as Plefka's expansion, and it takes about 1 page. See Plefka (1982). What is nice is that you can "see" where each term of F_{TAP} comes from: the zeroth-order term yields the "noninteracting" binary entropy terms; the first-order term yields the quadratic form; and the second-order term yields the Onsager correction terms. In effect, α is modulating the strength of the interactions while keeping the temperature and external field fixed.

Remark 1. *One odd thing about F_{TAP} is that it doesn't arise from a particular variational approximation to the measure $p(\sigma)$. In that sense, it's less interpretable than F_{mf}, which comes from the product measure approximation. From the point of view of the measure itself, is F_{TAP} essentially optimizing over joint measures that factorize into products of distributions $q_{ij}(\sigma_i, \sigma_j)$ over pairs of spins?*

Remark 2. *Notice that F_{TAP} is less than F_{mf}; thus the same relation holds between their minima. Is this intuitive?*

With this TAP free energy in hand, the natural question becomes: does it yield a good approximation to the mean of the SK measure? Or, more relevant to our purposes, is mm^T close to the Bayes-optimal estimator of xx^T? The answer is yes, and this is the main result of the paper Fan et al. (2021). The proof involves the Kac-Rice formula, to be discussed below. The moral of the story is: this justifies the use of the TAP free energy in Bayes-optimal estimation in the $\mathbb{Z}_2$ synchronization problem.

Remark 3. *In general, the variational objective is a good way to measure the correlations present in a high-dimensional measure. There are more sophisticated ways, including decompositions into convex combinations of approximate product measures. See Eldan (2020) and El Alaoui and Montanari (2022). These somehow relate to Stochastic Localization and sampling... Is it true that you can only sample efficiently if the Gibbs measure is approximately a single product measure?...*

12 Kac-Rice formula

How do Fan et al. (2021) prove that if m is a minimizer of F_{TAP}, then mm^T is close to the Bayes-optimal estimate? Essentially, just a counting argument. Obviously, the minimizers of F_{TAP} (and a priori there may be many, because this is a nonconvex function) are critical points of F_{TAP}. If we can count the number of critical points m such that mm^T is more than ϵ away from the Bayes-optimal estimate, and if we can show that this number goes to zero as n goes to infinity, we're done.

The canonical way to count critical points of a random function (in this case, F_{TAP}) is the Kac-Rice formula. The Kac-Rice formula says that the expected number of critical points of a high-dimensional random function f is the conditional expectation of the absolute value of the determinant of the Hessian of f, given that the gradient of f equals zero. In other words, if f typically has high curvature at its critical points, you expect many of them; whereas if f is typically flat near its critical points, you expect few of them.

Fan et al. (2021) actually apply Kac-Rice to F_{TAP} on constrained regions, because they only care about critical points below a certain threshold. Their main proposition is that the expected number of critical points in a region is at most $\exp(nS)$, where S is a rate function corresponding to the given region. If we can show that $S < 0$ for the region of interest, we are done. Since S is just

$$\frac{1}{n} \log \mathbb{E}[|\mathrm{Crit}(F_{TAP})|],$$

the analysis proceeds by some replica-like methods (interchanging the log and expectation), and using random matrix theory to study the expected determinant of the Hessian (which is essentially a perturbed GOE).

13 Stochastic localization and measure decomposition

Preface: we can talk about mean-field approximation of the free energy directly. To do this, start with the Gibbs variational principle, a variational formula for the free energy. Then only optimize over product distributions to obtain a lower bound mean-field approximation. See Eldan (2020).

We've seen that product measure approximations of Gibbs measures are nice, especially for computing the free energy.

Eldan (2020) asks the question: can we decompose a Gibbs measure μ as a mixture of (approximate) product measures μ_θ over some index set Θ?

Well, as El Alaoui and Montanari (2022) points out, if we had a way to sample from μ, then if $\theta \in \Theta$ denotes the realized randomness and $\sigma(\theta)$ denotes the random sample, then setting μ_θ to be the delta measure $\delta_{\sigma(\theta)}$ would do the trick.

But this is a little silly, because the entropy of the mixture is zero. Can we do this while (approximately) preserving the entropy? That is, can we ensure $\mathrm{Cov}\,[\mu_\theta]$ is typically approximately diagonal, and also that $\mathbb{E}_\theta H(\mu_\theta)$ is close to $H(\mu)$?

Eldan (2020) shows that the answer is yes, though there is a natural tradeoff between the two objectives. To do so, the author uses his stochastic localization (SL) process. Given a measure μ, the SL process associated to μ is a martingale $(\mu_t)_{t\geqslant 0}$ in the space of probability measures that starts at $\mu_0 = \mu$ and converges almost surely to $\mu_\infty = \delta_\sigma$, where σ is a random sample from μ. That is, SL provides a (theoretical) way to sample from μ. This gives us a way to fix our naive decomposition above. We run our for a certain (random) amount of time τ, and we let our decomposition be the mixture of $\mu_\tau(\theta)$, where $\theta \in \Theta$ is the realized randomness of the SL process. Entropy decreases along the SL process, but μ_τ becomes more of a product measure as τ increases. If we choose τ right, then this decomposition will achieve both our aims. (A simple proof is given in El Alaoui and Montanari (2022), add...)

With this result, Eldan (2020) can quantify the error of mean-field approximations to the free energy, add...

This builds on previous work of Eldan that shows that good decompositions of Gibbs measures can be obtained by conditioning on some S set of sites (so that θ is the random realization of the sites in S). (Is this intuitive? If you condition on the right set of sites, then you can split the sites into components that are roughly independent? add...)

A previous work Jain et al. (2019) studies free energy approximation via convex optimization. Instead of restricting the optimization set for the Gibbs variational principle, it relaxes it to optimizing over probability distributions on the hypercube, and uses techniques from SOS relaxations. add...

14 Optimization

Optimization of the Ising p-spin Hamiltonian in El Alaoui et al. (2021) (a very hard paper). Essentially a combinatorial optimization problem. The idea is to set up an efficient "IAMP scheme" (a fancy version of gradient descent), based on the Parisi formula for the minimum of the Hamiltonian. For this to yield a good approximation of the minimum, you need to make a "no overlap gap" assumption. Here, "overlap" refers to the normalized inner product $\langle \sigma^{(1)}, \sigma^{(2)} \rangle / N$ between two iid replicas drawn from the measure. The distribution of the overlap converges to some measure on $[0, 1]$ as $N \to \infty$. If this distribution has positive density on all of $[0, 1]$, then there is no overlap gap. But if the density vanishes on some subinterval $I \subseteq [0, 1]$, there will be a gap in the pdf. (Actually, it is shown in Gamarnik and Jagannath (2021) that an overlap gap is a barrier to efficient optimization. The overlap gap property (OGP) is known to occur in p-spin models for $p \geqslant 4$.) The actual details of this paper are very hard.

Remark 4. *Geometrically, if there is an overlap gap, then the Gibbs measure should look "bimodal", with at least two clusters?...*

Relation to the "complexity" of the loss landscape Auffinger et al. (2013), add...

15 Sampling

When is it easy/hard? See El Alaoui et al. (2022) for efficient sampling from the S-K model at high temperature $\beta < 1/2$, and a proof of hardness under some assumptions at low temperature $\beta > 1$. This paper is confusing because it uses stochastic localization to sample, which feels a bit circular... The main idea is that SL sampling can be done so long as you have good estimates of the Gibbs mean of μ_t, which is done as follows.

Computing the mean takes two steps. First run AMP to get a rough estimate of the Gibbs mean. This will land you in a region of strong convexity of F_{TAP}. So perform a few steps of gradient descent to F_{TAP} to improve the estimate.

Equivalent to diffusion models?, add...

Alternate methods of sampling: (Langevin) MC, score matching, diffusion models, normalizing flows, flow matching, Fisher-Rao/Wasserstein gradient flows; see the monograph Chewi (2023)...

16 Low-degree polynomials method

A perspective from Semerjian (2025): Working in a Bayesian setting, let $S \in \mathcal{S}$ denote the signal (distributed according to some prior) and let $Y \in \mathcal{Y}$ denote the noisy observation, where $\mathcal{S}, \mathcal{Y}$ are linear spaces. Suppose that a symmetry group G acts linearly on $\mathcal{S}$ and $\mathcal{Y}$ in such a way that the joint distribution (S, Y) is G-invariant. Then with respect to the squared error loss, the Bayes estimate $\hat{S}^{BO}(Y)$ of S (the posterior mean) is G-equivariant. If it is difficult to explicitly solve for $\hat{S}^{BO}(Y)$, this suggests the following natural *variational* approach: write

$$\hat{S}^{BO}(Y) = \min_{\hat{S}(Y) \text{ equivariant}} \mathbb{E}[d(S, \hat{S}(Y))|Y],$$

where the minimum is taken over the space of all equivariant estimates $\hat{S}(Y)$ of S, and restrict the minimization to a reasonable subspace of estimates. In particular, if restrict to equivariant polynomials in the entries of Y, then you obtain a version of the "low-degree polynomials method" from TCS. There are also connections to the Hunt-Stein theorem in statistics. (The Hunt-Stein theorem in the theory of hypothesis testing states that under some assumptions on the symmetry group G, a minimax test is necessarily invariant, so to find a minimax test it suffices to search over the space of invariant tests. The proof is by an averaging trick over the group G. In many problems invariance or equivariance of estimates or tests is desirable. For more on this, see Eaton and George (2021).)

(To verify that the Bayes estimate is G-equivariant, note that for any $y \in \mathcal{Y}$ and $g \in G$, if we write $\tilde{y} = gy$, then

$$\hat{S}^{BO}(gy) = \hat{S}^{BO}(\tilde{y}) := \mathbb{E}[S|Y = \tilde{y}] = g\mathbb{E}[g^{-1}S|g^{-1}Y = y] = g\hat{S}^{BO}(y),$$

where in the last step we used the fact that $(S,Y) \overset{d}{=} (g^{-1}S, g^{-1}Y)$. This g, g^{-1} gymnastics is the usual confusion with induced actions on functions out of a set...)

As a simple 1-dimensional example, if $Y = S + Z$ for $Z \sim N(0,1)$, then the joint distribution of (S,Y) is $\mathbb{Z}/2$-invariant under sign-flip, so that the equivariant low-degree polynomials are *odd* polynomials in Y.

17 Dynamical mean field theory

An iteration scheme to analyze gradient-based optimization algorithms. (Related to this, there seem to be recent works on SGD by Ben Arous and collaborators...)

18 Other models

In statistics, exponential random graph models (which include Erdos-Renyi and the edge-triangle model) are Gibbs measures with higher-order interactions, and to compute the MLE you need to approximate the partition function.

In Talagrand's books, he discusses the binary perceptron, Hopfield model, etc. These models might be of less interest to statistics.

19 Uncertainty quantification

The intersection with UQ is an emerging area, with a few papers like Clarté et al. (2023), add...

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